

ARYAN SINGH

Full Stack Software Engineer

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Summary

Full Stack Software Engineer with hands-on experience building AI-first products and scalable web applications from the ground up. Founding team member at an early-stage startup, owning end-to-end frontend architecture and AI-driven workflows. Strong expertise in React, TypeScript, Node.js, Python, and microservice-based systems, with deep exposure to real-time streaming (SSE), Retrieval-Augmented Generation (RAG) pipelines, and agentic AI systems. Passionate about shipping high-impact products in fast-paced environments.

Experience

Marxx AI

Sept 2025 – Present

Software Engineer

Gurugram, Haryana

- Led core product platform development as a founding team member, spanning ad research, ad creation, ideation testing, performance analytics, and AI-driven reporting workflows.
- Engineered a cross-platform **Chrome extension** (Manifest V3) for ad library ingestion, with a pluggable adapter/scrapper architecture, Shadow DOM-isolated React UI, and dynamic ad detection via MutationObserver to support Meta, TikTok, Google, and LinkedIn ad surfaces.
- Architected an agentic AI assistant with *Just-In-Time context retrieval*, using RAG-style context injection and session state management to drive contextual actions across search, report generation, entity intelligence, and ad creation flows.
- Created a hook-based action system with typed payload schemas and dynamic routing, enabling the AI agent to invoke platform workflows (search, analytics, report chat, test ad generation) via API endpoints and async job dispatching.
- Built semantic search over ad creatives using embedding-based retrieval and similarity scoring, supporting text and image queries with configurable filters and ranked result sets.
- Implemented streaming pipelines with Server-Sent Events (SSE) for token-level AI output delivery, using chunked processing and incremental response assembly across reports, chat, and ad analysis.
- Orchestrated queue-based AI ad generation with job scheduling, progress tracking, and streaming status via background task pipelines for image, video, and script-to-video creative workflows.

Recur Club

June 2024 – September 2025

Software Engineer Intern

Gurugram, Haryana

- Architected and built **Ask AICA**, an end-to-end AI investment chatbot, by laying the foundation of a new microservice, modeling data and configs, and assembling tool-orchestrated RAG pipelines; also delivered the complete frontend experience, improving time-to-answer by 65%.
- Shipped a Deal Status Tracking System with an interactive UI, reducing TAT by 27% and boosting deal visibility and closure rate.
- Launched an AI-generated company overview module that consolidated financials, key metrics, and business insights, giving users a 360° understanding and reducing research time by 72%.
- Engineered a streamlined NACH cancellation flow across vendors, reducing user effort by 82% and significantly enhancing user control and transparency.
- Crafted the frontend UI/UX for the AI-powered Magic Upload feature, enabling smart document tagging and boosting file retrieval speed by 79%.
- Refactored the client-facing frontend for the company onboarding flow using reusable components, enhancing UX consistency and driving a 40% increase in user engagement.
- Built an intuitive UI system for repayment refund tracking, streamlining operations and reducing repayment refund-related queries by 64%.

Open Source Contributions

google-gemini/gemini-cli | CLI · Merged May 2026

PR #25802

- Added a renderer-side post-processor that converts common LaTeX idioms in model output (arrows, set notation, Greek letters, fractions, sub/superscripts) into terminal-friendly Unicode, while preserving currency, Windows paths, regex escapes, and code spans. Shipped with 51 new unit assertions covering edge cases.

google-gemini/gemini-cli | CLI · Merged May 2026

PR #25827

- Fixed a bug where SessionStart hook **systemMessage** output rendered twice in the interactive UI by collapsing two emission paths into the event bus as the single source of truth, and broadened the gate to cover plain-text hooks — also surfacing previously-dropped BeforeAgent/BeforeTool plain-text messages.

vacanza/holidays | *Localization · Merged Apr 2026*

PR #3551

- Corrected the Arabic translation of the “(estimated)” / “(observed, estimated)” date labels, swapping the definite participle for the indefinite form so the parenthetical reads naturally. Cross-cut 77 files across 19 country modules, .po catalogs for multiple locales, and the matching test suite.

stdlib-js/stdlib | *Strings · Merged 2023*

PR #1173

- Contributed a new **string/base/altcase** utility to the stdlib strings namespace — converting input strings to alternating case — with full TypeScript typings, benchmarks, tests, REPL help, and documentation. Resolved a long-standing tracked feature request.

freeCodeCamp | *Curriculum · Merged 2025*

PRs #63789, #63723

- Fixed checkout lab tests to accept both implicit and explicit label associations and clarified lab instructions to reflect that either method is valid; also authored practical code examples for the Semantic HTML review page to improve learner clarity across each semantic element.

matrix.org | *Documentation · Merged Dec 2025*

PR #2981

- Merged the redundant *Cryptography* blog category into *Encryption* on the Matrix.org website, improving content discoverability and terminology consistency; contributed reasoned justification for the taxonomy decision during review.

Projects

Natak (Natak TV) | *Expo, React Native, react-native-webview, React Navigation, Google Sign-In, Expo Secure Store*

- Shipped the product as a mobile application by wrapping the existing web experience in an Expo-based native shell with a WebView host.
- Implemented native Google Sign-In with secure token storage and bridged the session into the WebView by syncing tokens into web localStorage, plus bidirectional logout signaling via postMessage for a consistent app session.

Cha-ai | *TypeScript, clsx, Vite, react-markdown, remark-gfm*

- Built a versatile, easily integrable React-based chatbot package with context-aware assistance, designed to embed into any host application with seamless integration and a customizable UI that matches the host platform.

Mudra | *React.js, Axios, dissu-talks, Vite, SCSS, Node.js, TypeScript, Express.js, MongoDB, Redis*

- Authored comprehensive integration documentation for Mudra, a secure and scalable SSO microservice for generating and validating JWT tokens across distributed ecosystems.
- Provided ready-to-use code samples and patterns enabling developers to implement robust authentication in distributed applications.

Guru Gang | *React, Vite, face-api.js, react-router-dom, Axios, dissu-talks, Gemini API, Node.js, MongoDB, CSS*

- Built an interactive learning platform offering courses, quizzes, assignments, and concept battles for students and teachers, with progress tracking and a distraction-free focus room.
- Integrated an AI-powered assistant (Gemini API) and face-api.js-based attention tracking to enhance the educational experience.

Skills

Languages: JavaScript, TypeScript, Python, Java, C++, C, SQL

Frontend: React.js, React Native, Redux, HTML5, CSS3, SCSS, Vite, Tailwind CSS

Backend: Node.js, Express.js, Spring Boot, REST APIs, Server-Sent Events (SSE), WebSockets

AI / ML: Retrieval-Augmented Generation (RAG), Agentic AI systems, LLM orchestration, Embedding-based semantic search, Prompt engineering

Databases: MongoDB, MySQL, Redis, PostgreSQL

Cloud & DevOps: AWS (EC2, ECS, S3, Lambda, Step Functions, SQS), Docker, Jenkins, CI/CD

Tools: Git, GitHub, Jira, Postman, Slack, Chrome DevTools, Manifest V3

Soft Skills: Cross-functional collaboration, Product ownership, Stakeholder communication, Problem-solving, Mentorship, Code review, Adaptability in fast-paced environments

Education

SRM University

Sep. 2021 – May 2025

Bachelor of Technology, Computer Science & Engineering

CGPA: 8.0

Positions of Responsibility & Achievements

Secured 2nd place at Vigyaan Hackathon.

Secured 3rd place at Startup Pitch-In.

Ranked in the Top 8 of 1,200 teams at Vihaan, a national-level hackathon.